KAMARUL FAHMI BIN KAMARUL NIZAM

Graduate Back-end, Cloud & DevOps Engineer | Kuala Lumpur, Malaysia | Available from Feb 2027
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EDUCATION

Bachelor of Software Engineering (Information System Development)
Universiti Kebangsaan Malaysia | Oct 2023 – Feb 2027 (expected)
CGPA 3.83 / 4.00 | MUET Band 4.5 | Yayasan TM full scholarship (since Oct 2024)

Quantum Research Camp
University of Tsukuba, Japan | Sep 2026
- Sessions at AIST and G-QuAT on superconducting qubits, quantum algorithms and RF testing.

TECHNICAL SKILLS

Cloud & DevOps: AWS (CDK, Lambda, API Gateway, DynamoDB, S3, CloudFront, Cognito, IAM, SSM), Google Cloud (Compute Engine, Pub/Sub, Cloud Tasks, IAM), Terraform, Infrastructure as Code, Docker, Docker Compose, nginx, CI/CD with GitHub Actions (OIDC), Linux, Bash, cron

Back end & data: Node.js, TypeScript, Fastify, Express, Laravel, REST APIs, WebSockets (Socket.IO), Prisma, Mongoose, MySQL, MongoDB Atlas

Front end & mobile: React, Next.js, Inertia, Tailwind CSS, React Native (Expo), Kotlin with Jetpack Compose

Languages: TypeScript, JavaScript, Python, PHP, SQL, Kotlin, Bash, C++ (competitive programming)

Tools & practices: Git, Nx, Changesets, pytest, Foundry, Vercel AI SDK, Gemini function calling, Agile, Scrum

WORK EXPERIENCE

Developer Intern, Versa
Fintech savings and investment app | Kuala Lumpur, on-site | Aug 2026 – Present
- Deployed the team's internal AI QA tool from a developer's laptop to serverless AWS with AWS CDK (TypeScript): Lambda, API Gateway (HTTP and WebSocket), DynamoDB, S3 and Cognito.
- Automated CI/CD deployments with GitHub Actions assuming an AWS IAM role over OIDC, so no long-lived AWS access keys are stored in GitHub.
- Fixed live-account AWS issues (IAM gaps, CORS preflight without a Lambda, WebSocket permissions, account-scoped S3 bucket names) and cut the main Lambda's memory from 1,536 MB to 512 MB.
- Writing Terraform (Infrastructure as Code) for a temporary, production-like Google Cloud load-test environment (Pub/Sub, IAM, Cloud Tasks, MongoDB Atlas restored from backup with personal data masked), plus a service-catalog tool that maps what each service publishes, subscribes to and calls (in progress).
- Extended the shared fund schema package (Mongoose, TypeScript), released it with Changesets as a pre-release then a versioned release, and served the new data from the fund-details API.
- Traced why eKYC would reject the new 2026 MyKad by studying the identity-verification provider's documentation, then added it to the supported document types (in code review).

Programming Mentor, Guru Muda UKM
Universiti Kebangsaan Malaysia | Oct 2024 – Aug 2025
- Mentored a cohort of 20 students for a year in programming fundamentals and problem-solving logic.
- Co-developed 12 learning modules with the faculty's academic committee.

PROJECTS

Haiwan Kita: volunteer-management platform | fahmikamarul.github.io/#project-hwk
Final-year project for Persatuan Haiwan Terbiar Malaysia (SAFM) | 2026 | Node.js, Fastify, TypeScript, Prisma, MySQL, Expo, Socket.IO, Docker, nginx
- Built the REST API and mobile app over 28 weeks with Agile, replacing the Excel sheets and WhatsApp groups SAFM used for volunteers; black-box and usability tested with SAFM's admins and volunteers.
- Implemented role-based access for 3 roles (volunteers, members, admins), live volunteer GPS on the admin map over WebSockets, and PDF certificates generated with headless Chromium.
- Containerised with Docker Compose behind nginx with TLS: private Docker networks, no database port on the host, a non-root API container and health checks.

KPZ Parcel: parcel-management system | github.com/FahmiKamarul/kpzparcel-server
Team of 5 | 2025 | Laravel 11, React, Inertia, Tailwind, MySQL, Kotlin, Jetpack Compose, Google Cloud

- Developed a Laravel REST API and web dashboard for counter managers and staff at the KPZ one-stop centre, and a native Android app for recipients.
- Delivered unique collection barcodes, a performance dashboard, and e-wallet or cash payments with PDF invoices; passed user acceptance testing with 3 testers.

Game server on Google Cloud | github.com/FahmiKamarul/palworld-server
Personal | 2026 | Docker Compose, Bash, cron, rclone, Google Cloud

- Hosted a dedicated game server on a Google Cloud VM and automated daily backups and restarts with cron, off-site backups to Google Drive with rclone, and Discord alerts.
- Scripted world import and co-op save migration, documented so the server can be rebuilt from a fresh VM.

Serverless app on AWS
Learning project | 2026 | AWS CDK, Lambda, API Gateway, S3, CloudFront, SSM, GitHub Actions, Nx

- Split the app into 4 AWS CDK stacks in an Nx monorepo; a GitHub Actions CI/CD pipeline assumes an AWS role over OIDC and deploys only changed stacks, back end first.
- Served the front end from private S3 behind CloudFront, reading the API URL from SSM at deploy time.

Sakuraya website
UKM Japan–Malaysia sociocultural centre | 2024 – 2025 | WordPress

- Created the first version in under 4 days while learning WordPress, worked with UKM's IT centre to get access to the university's WordPress platform, and kept the site updated until Aug 2025.

ADDITIONAL PROJECTS

- **MY Buddy** (3rd place, Generative AI Vibe Hackathon 2026; Next.js, Vercel AI SDK, Gemini): AI day-trip planner for Malaysia built in about 5 hours with a team of 4, using tool calling and property-based tests. github.com/FahmiKamarul/my-local-buddy-itinerary
- **AI Affordability Radar** (2025; Express, Gemini function calling, Google Places API): agent that returns the 10 closest homes a user can afford at a 50% debt-service ratio, from a 2025 Malaysian house-price dataset.
- **Marshall Assist** (Champion, Hackathon Za'ba 2025; LSTM): recognises airplane marshalling gestures from video and gives trainees real-time feedback; built in 6 hours with a team of 4.
- **On-chain Raffle** (2025; Solidity, Foundry, Chainlink): lottery on the Sepolia testnet with a verifiable random winner and automatic rounds; Foundry tests run in GitHub Actions CI. github.com/FahmiKamarul/lottery-smart-contract
- **YTM Claim Filler** (2026; Python, PyMuPDF, pytest): CLI that fills the monthly Yayasan TM claim form, working out working days around weekends and state public holidays.

ACHIEVEMENTS

- **3rd place**, Generative AI Vibe Hackathon 2026, FTSM UKM (May 2026)
- **9th of 140 teams**, Top Coders Malaysia 2025 national final; 1st among UKM teams (Nov 2025)
- **Champion**, Hackathon Za'ba 2025, UKM (May 2025)
- **1st runner-up**, Code Challenge 2025, UKM (May 2025)
- **2nd runner-up**, ProSolve National 2025, UiTM Shah Alam (Apr 2025)
- **Silver**, PPKOMP 2025, IIUM (Apr 2025)
- **9th, top 10 finalist**, CodeNection 2024, MMU (Dec 2024)
- **Bronze**, PPKOMP 2024 (May 2024)
- **Finalist**, Programming League National 2024, Universiti Malaya (May 2024)
- **4th place**, UM Game Jam 2024 (May 2024)
- **Finalist**, CodeNection 2023, MMU (Dec 2023)

CERTIFICATIONS

- IBM: Introduction to DevOps; Agile Development and Scrum (2026)
- Cisco: Introduction to Cybersecurity (2026)
- Google: Project Management Foundations; Data Analytics (4 courses); UX Design Foundations (2024 – 2026)